

Multivitamin–Multimineral Supplementation: When, Why, and for Whom?

A Physician's Perspective for Optimizing Patient Care

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Agenda



- **Introduction**
- **Who actually needs a MVMs?**
- **When can supplementation may harm?**
- **How can physicians and pharmacists collaborate?**
- **Conclusion**

Multivitamin-minerals Supplements: Facts

- The supplemental vitamin industry is **worth \$39 billion USD and continues to grow.**
 - Nearly **half of adults** in the U.S. and **70% of older adults ages 71+** take a vitamin.
 - About **one-third** of them use a **comprehensive multivitamin pill.**

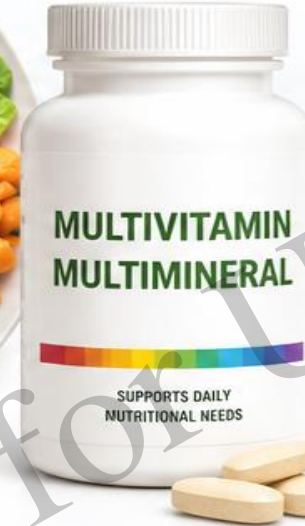
[Should I Take a Daily Multivitamin? | The Nutrition Source | Harvard T.H. Chan School of Public Health](#)

“

For those who eat a healthful diet, a multivitamin may have **little or no benefit.**



HARVARD
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SCHOOL OF PUBLIC HEALTH



“

If you follow a healthy diet, you can get all of the vitamins & minerals you need **from food.**



JOHNS HOPKINS
MEDICINE

»»» **BUT...**

In a busy,
STRESSFUL LIFE...
do we really eat
a healthful diet?



**LONG
WORK HOURS**



**ON-THE-GO
LIFESTYLE**



**FAMILY
RESPONSIBILITIES**



**CONVENIENCE
FOODS**



STRESS



TIME PRESSURES



POOR FOOD CHOICES



The gap between what we know and what we do can lead to **hidden nutritional gaps.**



When a MVM May Be Recommended?

Development of Nutritional Deficiency

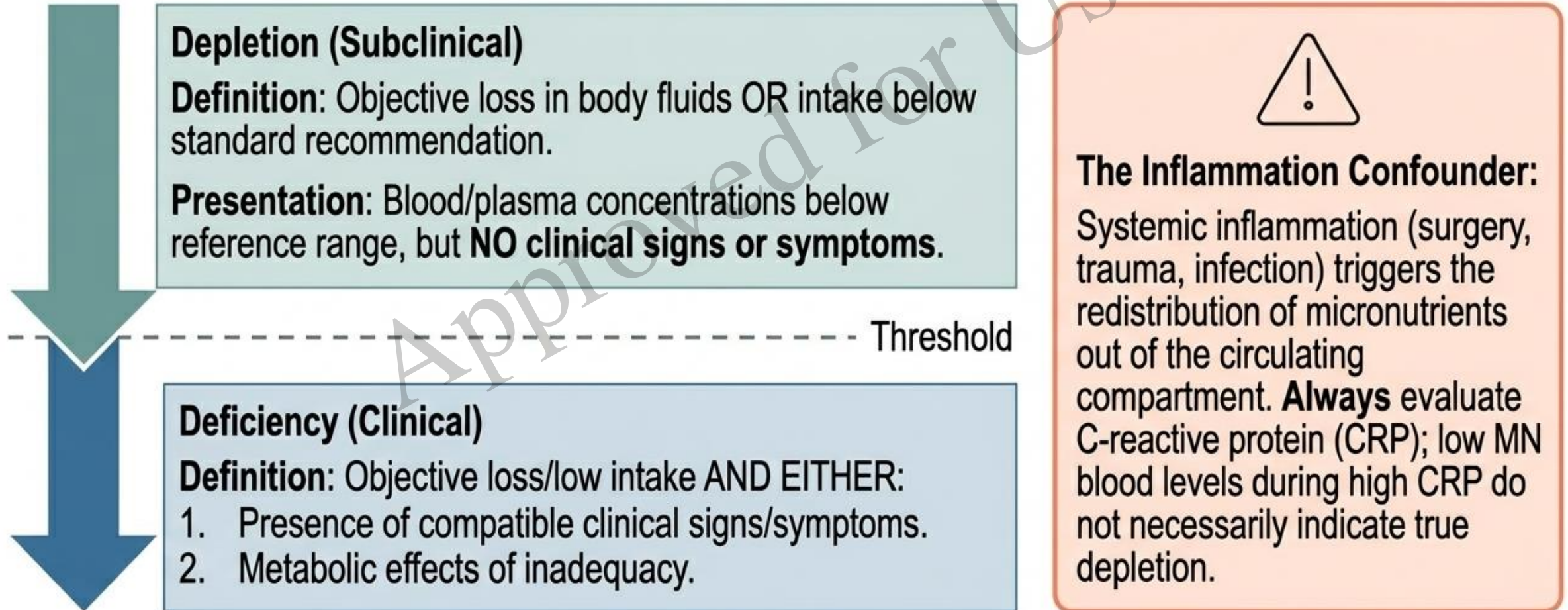
- **States of nutritional deficiency or excess occur when nutrient intake is not balanced with specific requirements for optimal health.**

When micronutrient intakes are inadequate, **suboptimal cellular/physiological functions** can occur in advance of developing a **classic symptomatic deficiency condition**

<https://www.hsph.harvard.edu/nutritionsource/multivitamin/>.

ESPEN Guidelines: Precision in Micronutrient Assessment

Low blood levels do not always equal clinical deficiency. Precision in terminology guides appropriate supplementation.



MVM: Is it truly a necessity?

Use of multivitamins & minerals supplements has been shown to improve micronutrient status among adults & children

(Bailey et al, 2012; Murphy et al, 2007; NIH, 2006)

Populations Potentially at Risk for Nutrient Deficiencies

<u>At-Risk Population or Life Cycle Stage</u>	<u>Nutrients of Concern that Could Potentially be Corrected by Supplementation</u>
Those living in poverty (especially children)	Iron, calcium, magnesium, folate, vitamins A, B ₆ , C, D, and E
Women taking oral contraceptives	Zinc, folic acid, B ₆ , and B ₁₂
Adolescent women	Iron and calcium
Pregnant women	Iron and folic acid
Elders	B ₁₂ and vitamin D, multiple micronutrients
Vegans	B ₁₂
Those following a calorie-restricted, allergy-restricted diet	Multiple nutrients
People with dehydrated	Vitamin D

Clinicians should be aware of these at-risk subgroups & **complete a nutrition assessment to determine need for supplementation on an individual basis if** nutrition status cannot be improved by dietary changes alone.

Higher-Risk Groups: The Elderly

The elderly are at risk for poor food intake

- Difficulty chewing & swallowing food
- Experiencing unpleasant taste changes caused by multiple medications
- Isolation & loneliness that can depress appetite.
- Impaired absorption of vitamin B12. (*The National Academy of Medicine recommends that people over the age of 50 eat foods fortified with vitamin B12 or take vitamin B12 supplements that are better absorbed than from food sources.*)

<https://www.hsph.harvard.edu/nutritionsource/multivitamin/>.

Institute of Medicine. Food and Nutrition Board. Dietary Reference Intakes: Thiamin, Riboflavin, Nicotinic Acid, Vitamin B6, Folate, Vitamin B12, Pantothenic Acid, Biotin, and Choline. Washington, DC: National Academy Press, 1998.

Higher-Risk Groups: Malabsorption Conditions

Any condition **that interferes with normal digestion can increase risk of poor absorption of one or several nutrients:**

- Celiac Disease, ulcerative colitis, or cystic fibrosis.
- Surgeries: bariatric surgery & Whipple procedure that involves many digestive organs.
- Illnesses associated with vomiting or diarrhea can prevent nutrients from being absorbed.
- Alcoholism can prevent nutrients, including several B vitamins & vitamin C, from being absorbed.

Higher-Risk Groups: **Certain Medications**

- **Some diuretics:** can deplete the body's stores of magnesium, potassium, & calcium.
- **Proton pump inhibitors:** may prevent absorption of vitamin B12 and possibly calcium and magnesium.
- **Levodopa & carbidopa** prescribed for Parkinson's disease can reduce absorption of B vitamins including folate, B6, & B12.



MVM Supplementation : The Evidence

- Female infertility treatment has been shown to improve with micronutrient supplementation, particularly with combinations of folic acid; vitamins B6, C, and D; iodine; selenium; iron; and/or omega-3 fatty acids.
- In a small pilot study, MVMS have shown better results versus folic acid alone in ovulation induction among women undergoing fertility treatment.

Modest effects of dietary supplements during the COVID-19 pandemic: insights from 445 850 users of the COVID-19 Symptom Study app

Panayiotis Louca ¹, Benjamin Murray ², Kerstin Klaser ²,
Mark S Graham ², Mohsen Mazidi ¹, Emily R Leeming ¹,
Ellen Thompson ¹, Ruth Bowyer ¹, David A Drew ³, Long H Nguyen ³,
Jordi Merino ³, Maria Gomez ⁴, Olatz Mompeo ¹, Ricardo Costeira ¹,
Carole H Sudre ⁵, Rachel Gibson ⁶, Claire J Steves ¹, Jonathan Wolf ⁷,
Paul W Franks ⁴, Sebastien Ourselin ², Andrew T Chan ³,
Sarah E Berry ⁶, Ana M Valdes ^{1,8}, Philip C Calder ⁹, Tim D Spector ¹,
Cristina Menni ¹

- Results In 372 720 UK participants (175 652 supplement users and 197 068 non-users), those taking probiotics, omega-3 fatty acids, multivitamins or vitamin D had a lower risk of SARS-CoV-2 infection by 14% (95% CI (8% to 19%)), 12% (95% CI (8% to 16%)), 13% (95% CI (10% to 16%)) and 9% (95% CI (6% to 12%)), respectively, after adjusting for potential confounders.
- No effect was observed for those taking vitamin C, zinc or garlic supplements.
- On stratification by sex, age and body mass index (BMI), the protective associations in individuals taking probiotics, omega-3 fatty acids, multivitamins and vitamin D were observed in females across all ages and BMI groups, but were not seen in men.
- The same overall pattern of association was observed in both the US and Swedish cohorts

The Cognitive Frontier: A Paradigm Shift

While early meta-analyses (including Cochrane reviews and PHS II) showed mixed or null effects on cognition, recent robust data from the COSMOS ancillary studies demonstrate significant cognitive preservation in older adults (>60 yrs) taking daily MVMs for 1 to 3 years.



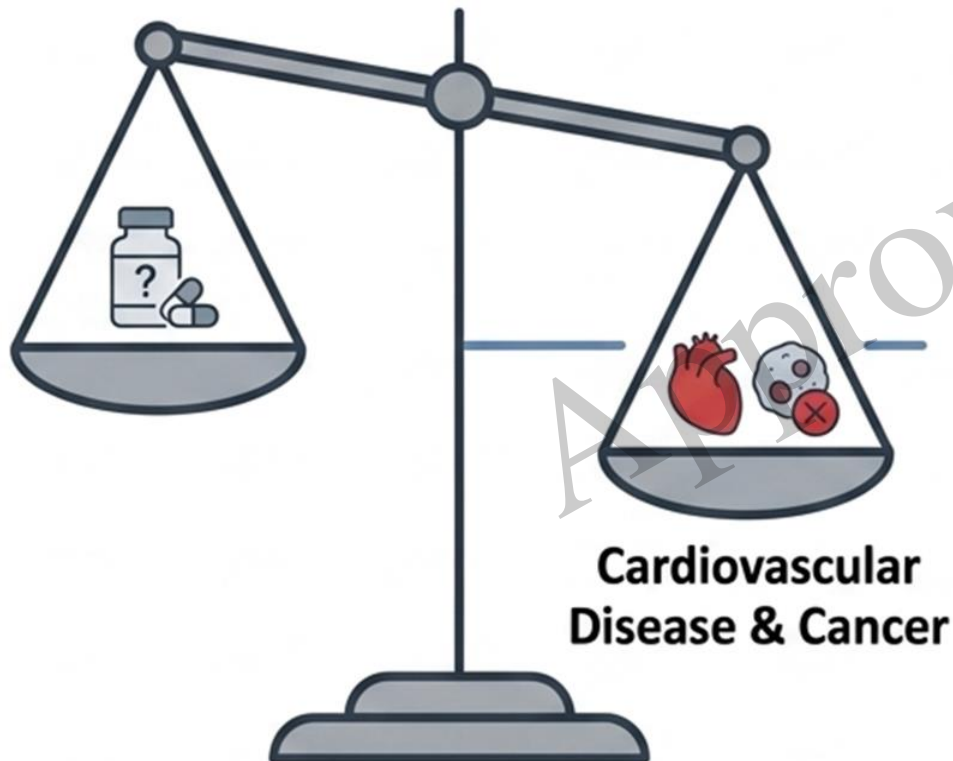
COSMOS-Mind (n=2,262): Improved global cognition (recall, recognition, learning). Benefits were most pronounced in patients with a history of CVD.

COSMOS-Web (n=3,562): Significantly better memory performance at 1 year and sustained across 3 years of follow-up.

COSMOS-Clinic (n=573) & Meta-analysis: Confirmed improvements in episodic memory and global cognition over time across >5,000 pooled participants.

Primary Prevention: The Baseline Evidence

Extensive systematic reviews confirm that for the general, well-nourished population, routine MVM supplementation does not reliably prevent primary cardiovascular disease or overall cancer mortality.



USPSTF Recommendations (2022 Update):



Against: Beta-carotene or Vitamin E supplements for the prevention of CVD or cancer (Grade D).



Insufficient Evidence: Multivitamins, or single/paired nutrients (other than beta-carotene/Vit E) for the prevention of CVD or cancer (Grade I).



The Use of Multivitamin/Multimineral Supplements: A Modified Delphi Consensus Panel Report

Jeffrey B. Blumberg, PhD¹; Hellas Cena, MD, PhD²; Susan I. Barr, PhD³; Hans Konrad Biesalski, MD⁴; Ricardo Uauy Dagach, PhD⁵; Brendan Delaney, MD⁶; Balz Frei, PhD⁷; Manuel Ignacio Moreno González, MD, MSc⁵; Nahla Hwalla, PhD, RD⁸; Ronette Lategan-Potgieter, PhD⁹; Helene McNulty, PhD, RD¹⁰; Jolieke C. van der Pols, PhD¹¹; Pattanee Winichagoon, PhD¹²; and Duo Li, PhD¹³

Based on current knowledge, long-term use of MVMS with an amount not exceeding UL is safe in healthy adults

Navigating the Unregulated MVM Landscape

Because MVMs have no standard regulatory definition, clinicians must differentiate product classes to match patient needs.



1. Basic / Broad Spectrum

Profile: Once-daily; contains most vitamins/minerals close to Recommended Dietary Allowances (RDAs) / Daily Values (DVs).

Clinical Use: General nutritional insurance; avoiding exceeding Tolerable Upper Intake Levels (UL).



2. High Potency

Profile: Contains amounts substantially higher than DV/RDA, sometimes exceeding UL. Often multi-pill packs.

Clinical Use: Targeted correction of confirmed deficiencies (e.g., post-bariatric surgery, malabsorption syndromes).



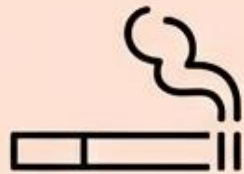
3. Specialized / Condition-Specific

Profile: Basic MVMs combined with botanicals, probiotics, or specific nutrient clusters (e.g., AREDS formulas for macular degeneration).

Clinical Use: Highly specific patient profiles; requires careful screening for herb-drug interactions.

Recognizing Risks, Toxicities, and Interactions

Basic MVMs are generally safe, but specific contexts present significant risks for harm:



Beta-Carotene in Smokers:

High-dose beta-carotene (and Vitamin A) is actively contraindicated current and former smokers due to a demonstrated increased risk of lung cancer incidence and mortality.



Drug-Nutrient Interactions:

Vitamin K variations in MVMs can critically alter the efficacy of warfarin and other anticoagulants. Patients must maintain highly consistent daily intakes.



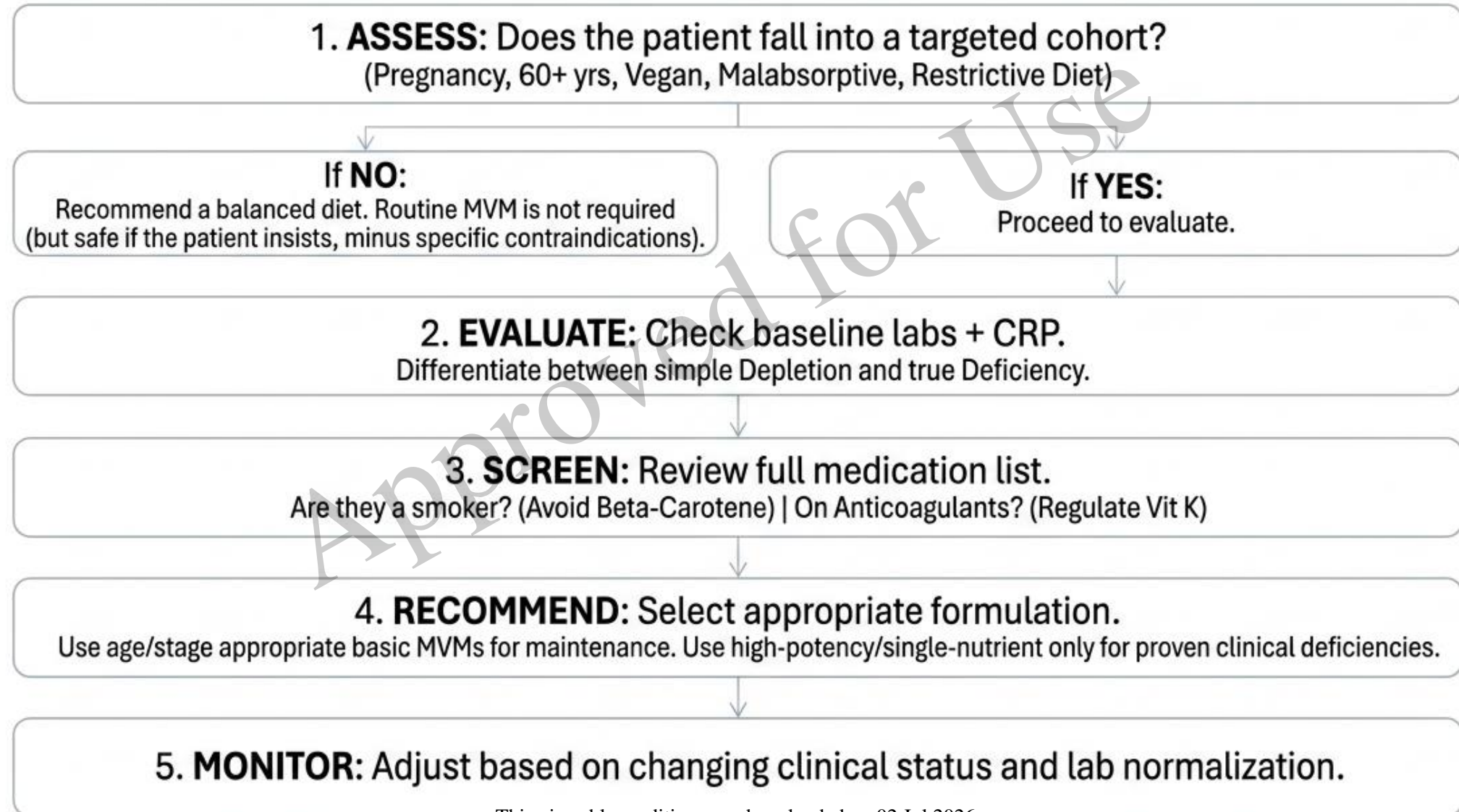
Masked Deficiencies: High-dose folic acid ($>1,000 \mu\text{g}/\text{day}$) can mask the hematologic signs of a Vitamin B12 deficiency, allowing irreversible neurological damage to progress undetected.



Excessive Cumulative Intakes:

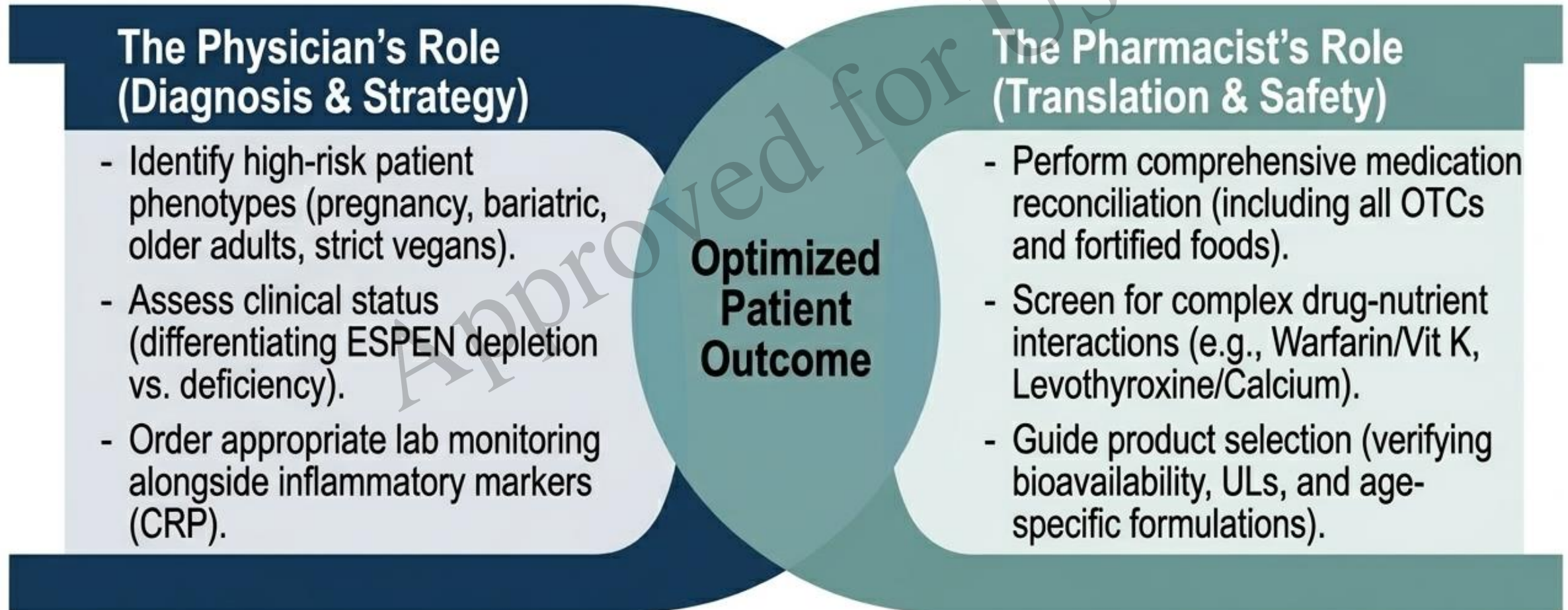
Patients combining MVMs with highly fortified foods or single-nutrient supplements risk exceeding the Tolerable Upper Intake Level (UL) for nutrients like iron (hemochromatosis).

A Practical Approach to MVM Supplementation



Physician-Pharmacist Synergy in Supplementation

With thousands of unregulated MVM products on the market, interprofessional collaboration is essential to transition patients from self-prescribed fads to evidence-based medical nutrition therapy.



In conclusion...Remember that:

- The main purpose of a multivitamin is to **fill in nutritional gaps** & provides only a hint of vast array of healthful nutrients & chemicals naturally found in food.
- ✓ Food first—but recognize when diet alone is insufficient..
- ✓ **Choose quality products with evidence-based formulations & avoid unnecessary mega doses.**
- **Interprofessional collaboration transforms MVM supplementation from product recommendation into evidence-based patient care.**

Thank You!

The **right patient**, the **right supplement**,
the **right dose**—and the **right healthcare team**.



RIGHT PATIENT

Identify need,
assess risk



RIGHT SUPPLEMENT

Evidence-based,
quality assured



RIGHT DOSE

Individualized,
safe & effective



RIGHT HEALTHCARE TEAM

Physician–Pharmacist collaboration
for optimal outcomes



Better nutrition. Better care. Better outcomes—*together.*

Directed to healthcare professionals
Promotion code: PM-EG-CNT-26-00012
preparation date: June 2026
Adverse event should be reported to:
Mystory.eg@haleon.com